The empirical recurrence for OEIS A203176, proved by transfer matrix

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Abstract

OEIS A203176 counts arrays of width 3 over $\{0, 1, 2\}$ under local conditions relating each cell to its neighbours or to the cells preceding it. The entry carries an empirical recurrence of order 9 contributed by R. H. Hardin. It is true, and it is decidable rather than empirical: every condition is settled inside three consecutive array rows, so the count is a walk count on $S = 36$ states.

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1 The conjecture

OEIS A203176 is "Number of n X 3 0..2 arrays with every 1 immediately preceded by 0 to the left or above, no 0 immediately preceded by a 0, and every 2 immediately preceded by 0 1 to the left or above.". It has offset 1 and begins

$$2, 2, 7, 17, 41, 97, 235, 607, \ \dots$$

The entry states, as a conjecture contributed by its author and never marked settled:

> Empirical: a(n) = 2*a(n-2) +10*a(n-3) +5*a(n-4) -a(n-5) -31*a(n-6) -7*a(n-7) +30*a(n-9) for n>10.

As of the "Last modified" line on the live entry (Jun 14 2014 17:24:20, revision 11) this is still recorded as empirical, and nothing on the entry records it as proved.

2 What is being counted

The array has 3 cells across, with entries in $\{0, 1, 2\}$. The NEIGHBOURS of a cell are the cells immediately to its left, to its right, above it and below it, those that lie inside the array. The PREDECESSORS of a cell are the cell immediately to its left and the cell immediately above it, those that lie inside the array. A cell in the top left corner has none at all, so a clause demanding a predecessor of a given value forbids that value there — which is what makes the first published term what it is. The entry asks that

- every cell carrying 1 has at least one predecessor carrying 0;
- no cell carrying 0 has a predecessor carrying 0;
- every cell carrying 2 has the two cells to its left, or the two above it, reading 0 then 1 in that order;

and nothing else.

3 Three rows decide everything

Lemma 1. *Every clause above is decided by three consecutive array rows. An adjacency clause for a cell of row i reads rows $i-1$, i and $i+1$; a precedence clause for a cell of row i reads rows $i-2$, $i-1$ and i, and columns at most two to the left.*

Proof. The neighbours of a cell lie one step away in each of the four directions, and the predecessors one step to the left or one step up; a clause naming two cells in a row or column reaches two steps, never more. □

So the state is the pair (row above, current row), with a row not yet written carried as $\perp$. A step appends a row: the precedence clauses for the NEW row are settled at once, because everything they read is already written, and the adjacency clauses for the MIDDLE row of the window are settled because the row below it has just arrived. Every clause here reaches backwards only — to the left and upwards — so a row is settled as it is written and every state may end an array; the end vector is the all-ones vector. With M the adjacency matrix of the digraph on those states,

$$a(n) \;=\; \iota^{\top} M^{\,j} \tau, \qquad S = 36.$$

The walk index j and the entry's own n differ by a constant, read off by matching the published terms rather than assumed. In particular a is C-finite.

4 The proof

Lemma 2. *Let $q(t) = t^9 - \sum_i c_i t^{9-i}$ be the characteristic polynomial of the conjectured recurrence. The residual $a(n) - \sum_i c_i a(n-i)$ equals $\iota^{\top} M^{\,j} q(M)\tau$ for the corresponding walk index j, so the recurrence holds from some point on if and only if $\iota^{\top} M^j q(M)\tau = 0$ for all large j, and it is enough to examine $j < S$.*

Proof. Factor M^j out of $M^{j+9} - \sum_i c_i M^{j+9-i}$. For the bound, M^S is an integer combination of $I, M, \ldots, M^{S-1}$ by Cayley–Hamilton, so the numbers $u_j = \iota^{\top} M^j q(M)\tau$ obey the monic recurrence given by the characteristic polynomial of M; S consecutive zeros force every later one, and the last nonzero u_j pins the threshold exactly. □

Theorem 3.

$$a(n) \;=\; 2\,a(n-2) + 10\,a(n-3) + 5\,a(n-4) - a(n-5) - 31\,a(n-6) - 7\,a(n-7) + 30\,a(n-9)$$

for every $n > 10$.

Proof. The vector $w = q(M)\tau$ was formed by 9 matrix–vector products in exact integer arithmetic and $\iota^{\top} M^j w$ evaluated for $j = 0, 1, \ldots,$ again exactly; those integers vanish from the index corresponding to $n = 10 + 1$ onwards and the run continued until the working vector was identically zero. □

5 Verification

Four checks, each independent of the algebra above.

First, the model reproduces all 29 terms the entry publishes, exactly, in integer arithmetic.

Second, the count was recomputed for the smallest arrays straight from the definition: enumerate every array of that size in the orientation the entry's own name writes and test each clause on each cell directly. No window, no state and no transfer matrix are involved, and the published terms came back.

Third, the conjectured recurrence was evaluated directly on the published terms, with no matrices involved, and holds wherever the proved range and the data overlap.

Fourth, the annihilation test of Lemma 2 was carried out in exact integer arithmetic on the whole vector rather than on a sampled prefix.

References

[1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, https://oeis.org, 2026. Sequence A203176.

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